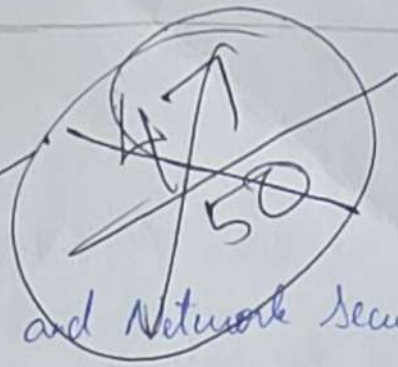


Analytical Questions:

Good



Sh

Name: Enad Jarroqui

Reg No: 192511019

Course Code / Name: CSA-5104 / Cryptography and Network Security

1. Key 3

Decryption:

Cipher Text: PHHW	PH	DIWHU	WKH	WRJD	SDUWB
Plain Text: MEET	ME	AFTER	THE	TOGA	PARTY

2. Plaintext: SECURITY

Formula:

$$C = (P + K) \bmod 26$$

Cipher Text: DPN FCTEJ

3. * Caesar Cipher has only 25 possible keys. An attacker simply tries every shift until meaningful text appears.

* This is called brute force attack.

* Time Complexity:

$$O(26) \approx O(1)$$

4. * Although a monoalphabetic has:

$26! \approx 4 \times 10^{26}$ possible keys, each plaintext letter is always replaced by the same ciphertext letter.

* Hence frequency analysis can break the cipher despite its huge key space.

5. Atbash Cipher:

Decrypting:

THIS IS A SECRET

6. Analytical Approach:

1. Count the frequency of every ciphertext letter.
2. Since E is the most common letter in English
3. Substitute this mapping throughout the ciphertext.
4. Look for recognizable words.
5. This iterative process gradually reveals the plaintext.

7. In a polyalphabetic cipher, one plaintext letter can encrypt into different ciphertext letters depending on the key.

Eg: Plaintext E may become

L, Q, M, K

Therefore, letter frequencies become spread out.

8. Key: KEY

Plaintext: CRYPTOGRAPHY

Key: KEYKEYKEYKEY

Using, $C = (P + K) \bmod 26$

Ciphertext: MUWZXMQUYZLW

9. Key: LEMONLEMONCE

$\Rightarrow P = (C - K) \bmod 26$

Plaintext: ATTACKATDAWN

10. When a short keyword is repeated, the encryption pattern also repeats.

* Repeated patterns in the ciphertext reveal the key length.

An attacker can use Kasiski Examination:

1. Find repeated ciphertext sequences
2. Measure distances between them
3. Estimate key length

Therefore, repeating a short keyword makes the Vigenère cipher vulnerable to cryptanalysis.

11. When the Vigenère key is equal in length to the plaintext, completely random, and used only once, it becomes a One-Time Pad (OTP).

Security:

* Every plaintext character is encrypted using a unique random key character.

* There are no repeating patterns.

12.

Feature	Monoalphabetic	Polyalphabetic
Alphabet Used	One substitution alphabet	Multiple substitution
Letter Mapping	Fixed	Changes with key pos
Frequency Pattern	Preserved	Hidden
Entropy	Lower	Higher

13. Letters are represented as numbers:

$$A=0, B=1, \dots, Z=25$$

Formula:

$$C = (P + K) \bmod 26$$

$$\text{Plaintext} = C(2)$$

$$\text{Key} = 5$$

$$(2+5) \bmod 26 = 7$$

$$\text{Ciphertext} = H$$

14. Key Matrix:

$$K = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$$

$$\text{Plaintext: } H=7, I=8$$

$$\text{Vector: } P = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

$$KP = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 23 \\ 61 \end{bmatrix}$$

$$61 \bmod 26 = 9 \rightarrow J$$

$$\text{Ciphertext: } XJ$$

15. $\text{Key: } \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}$

Determinant

$$(2 \times 4) - (4 \times 2) = 0$$

Since the determinant is 0, the matrix has no inverse.

Hence, encryption/decryption fails.

16. The determinant must satisfy:

$$\gcd(\det(K), 26) = 1$$

This guarantees that the determinant has a modular inverse.

If no modular inverse exists:

- & The key matrix cannot be inverted

- & Decryption becomes impossible.

$\therefore$, the determinant must be relatively prime to 26

17. Hill cipher encrypts multiple letters together using matrix multiplication.

Advantages:

- & One plaintext letter affects several ciphertext letters.

- & Strong diffusion.

- & Letter frequencies are hidden.

Thus, linear algebra significantly increases security.

18. A nearly uniform ciphertext frequency suggests a Polyalphabetic Cipher, such as the Vigenère Cipher.

Reason:

- & The same plaintext letter encrypts to different ciphertext letters.

- & Frequencies are spread across many letters.

19. Example Substitution:

Plain: ABCDEFGHIJKLMNOPQRSTUVWXYZ
Cipher: QWERTYUIOPASDFGHJKLZXCVBNM

Secure key sharing:

- ✧ Exchange key using a secure channel
- ✧ Encrypt the key with public-key cryptography.
- ✧ Never transmit the key in plaintext.

20. Yes, If the same plaintext block is encrypted using same key matrix,

$C = K P \pmod{26}$, the output ciphertext block will also be identical.

21. No, A simple Caesar cipher has the form:

$$C = (P + K) \pmod{26}$$

The given equation:

$C = (7P + 3) \pmod{26}$ includes multiplication and addition, making it an Affine Cipher, not a Caesar cipher.

22. In a known-plaintext attack, the attacker knows some plaintext and its corresponding ciphertext.

Steps:

1. Match plaintext letters with ciphertext letters
2. Determine substitution mappings
3. Recover additional mappings.

6. The Vigenère cipher uses different substitution alphabets, creating strong confusion by hiding the relationship between plaintext and ciphertext.

24. "The" is the most common English word.
An attacker searches for repeated three-letter ciphertext patterns. If one pattern appears frequently, it is likely to represent "The".

- 25.
- Caesar Cipher: Only 25 keys, easily broken by brute force.
 - Monoalphabetic Cipher: Large key space but vulnerable to frequency analysis.
 - Vigenère Cipher: Uses multiple alphabets, reducing frequency analysis effectiveness.
 - Hill Cipher: Uses matrix operations and provides strong diffusion, making cryptanalysis more difficult.

26. Message:
RETREAT IMMEDIATELY
Ciphertext received
RRMTEEIATETMLOYHE
The ciphertext is not correct.
In a proper 3-rail Fence Cipher:
• Characters must follow a zig-zag pattern across the three rails.
• The intercepted ciphertext indicates that the rail traversal or zig-zag order was applied incorrectly.

27. Cipher text:

TKAATNACDTWA

Known plaintext:

ATTACK AT DAWN

No padding is required

28. If a columnar transposition key contains repeated letters, assigning column numbers becomes ambiguous.

This may cause:

- * Different encryption outputs.

- * Incorrect decryption if sender and receiver use different ranking rules.

29. Plaintext:

SECRET MISSION

Ciphertext:

SCEITNERTSSMO

For a correct 2-rail Fence cipher:

- * Characters should alternate between the two rails in a zig-zag pattern.

30. Key lengths:

- * First Key = 4, Second Key = 6

Possible matrix dimensions:

- * 4×6 , 6×4

Since, $4 \times 6 = 24$, the ciphertext fits the matrix exactly.

- * No padding required.

Key: ZEBRA

Letter: A B E R Z

Og Pos: 5 3 2 4 1

Col Order: 1 2 3 4 5

Analytical Order:

& Arrange the ciphertext into columns according to the alphabetical order of key.

& Restore the Og column pos.

& Read the table row-wise to recover plaintext.

32. A transposition cipher does not change letters, it only rearrange their position.

Therefore, a single-letter frequencies remain identical to plaintext.

33. This can happen why:

& Different keys generate the same column permutation.

& The plaintext contains repeated characters or repeated blocks

& The permutation has symmetrical properties.

34. Given: Ciphertext length = 21

Systematic Method:

1) Assume 2 rails and decrypt.

2) Check if the result is meaningful English

3) Repeat for 3, 4, 5 ... rails.

4) Stop when the decrypted msg becomes readable.

35. Each key has:
 $5! = 120$ possible permutations.

Total key combinations:
 $120 \times 120 = 14,400$

A larger permutation space significantly increases resistance to brute-force attacks compared with a single transposition cipher.

36. Using a fixed padding such as 'X' makes the padding easy to recognize.

An attacker can:

- Identify the message length
- Remove predictable padding
- Infer structural info about the plaintext.

37. Ciphertext length is 17 and since 17 is prime, the only possible table dimensions are:

& 1×17

& 17×1

Therefore, very few column arrangements are possible. Many common transposition matrix sizes are impossible.

38. If an attacker assumes the plaintext was filled row-wise instead of column-wise

& columns are reconstructed incorrectly.

& Characters appear in the wrong positions.

39. In a 3-rail fence cipher:

& The first character always starts on the top rail

& The last character often ends on the top or bottom rail depending on message length.

10. No. of rows:

$$\left\lceil \frac{23}{6} \right\rceil = 4$$

Total cells:

$$4 \times 6 = 24$$

Empty cells:

$$24 - 23 = 1$$

41. Possible permutations:

$$5! = 120$$

Total unknown combinations:

$$120 \times 120 = 14,400$$

42. Analytical approach:

& Attempt decryption using a single transposition

& If the output remains unreadable, suspect double transposition

43. Repeated letters create ambiguity in assigning column numbers.
Eg: BALLOON

The two L's and two O's require a consistent ordering rule.
Without an agreed convention:

& Encryption and decryption produce different results.

44. A regular repetition every 5 letters is more likely to indicate a Columnar Transposition cipher.

Reason:

& The repetition usually corresponds to no. of columns or key length.

45. If the ciphertext characters are distributed almost equally among the rows, the most likely no. of rails is 3.

Reason:

& These rails commonly produce balanced row length for many plaintext sizes.

46. When one key length is a factor of the other:
* Certain transposition patterns may repeat.
* The second transposition may partially undo or reinforce the first.
47. Transposition itself does not change the no. of characters.
The ciphertext becomes longer only when:
* Padding characters are added to complete the final row or ^{col.} ~~the~~.
* Plaintext length is not a multiple of key length.
48. Possible key permutations:
 $6! = 720$
An attacker must try up to 720 different col orders.
49. Plaintext: CRYPTOGRAPHY
Analysis:
Compare the ciphertext with the expected outputs for 2-rail and 3-rail ⁱ fence encryption.
Therefore, the ciphertext corresponds to 2-rail fence encryption.
50. Advantages:
* No ambiguity from repeated letters
* Easier to assign col order
* Faster implementation
Disadvantages:
* Harder for users to remember
* Random numeric keys may be difficult to communicate securely.
* Less mnemonic than meaningful words.